20190822
first draft release to internal

20190827

added LNA Power BUCK_1V8 on module symbol
change active antenna power option form Jumper to 0ohm resistor
added RF Coaxis Connector in RF Path

20190829
modify module symbol

20190904
R551 replace J504

20190905
Separe VAUXADC/VRTC Power source

20190916

26M TCXO follow 3333P parts
added Chip Antenna
remove 105 degree SAW parts for improve layout
added C530/C531/C532/C533 for pi-type matching
L502 change to 3.9nH

20190917
modify unmount symbol

20190918
added C427~429 for Control pin

20190921
SOC cap < 1uf change to C0201

20191001
modify module pin defined

20191107
vcore power source change to external

蹈 節 蒿 M T 綜 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 楾 λ カ 蒿 厩 堕 跋 縲 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title
History

Size D
Document Number
AG3335M EVK V2
Rev 0.5

Date: Monday, March 23, 2020

20190819

蹈 笷 蒚 M T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 檋 λ カ 荂 厫 隉 跋 繠 ︿ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title
Ball Map

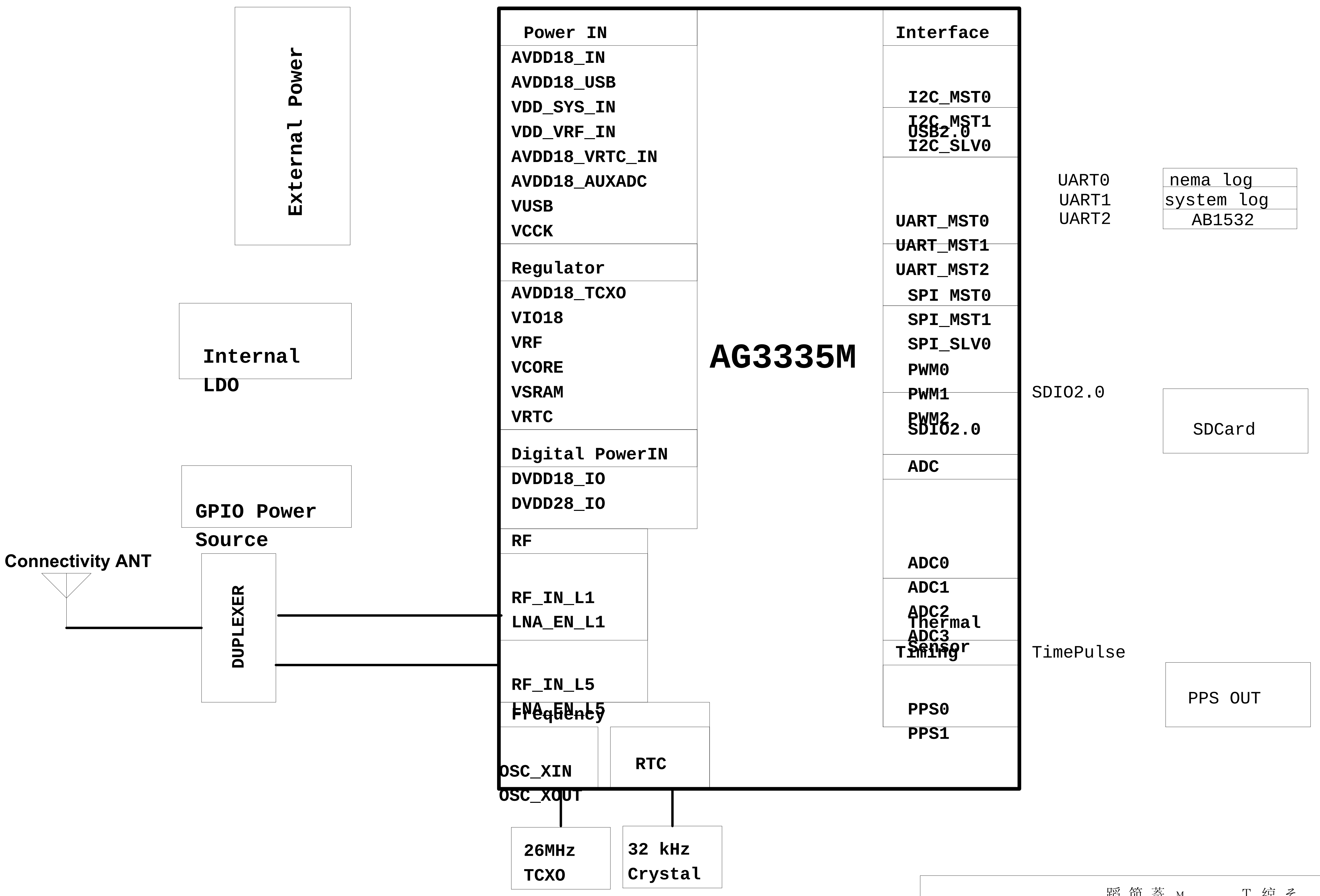
Size
D

Document Number
AG3335M EVK V2

Rev
0.5

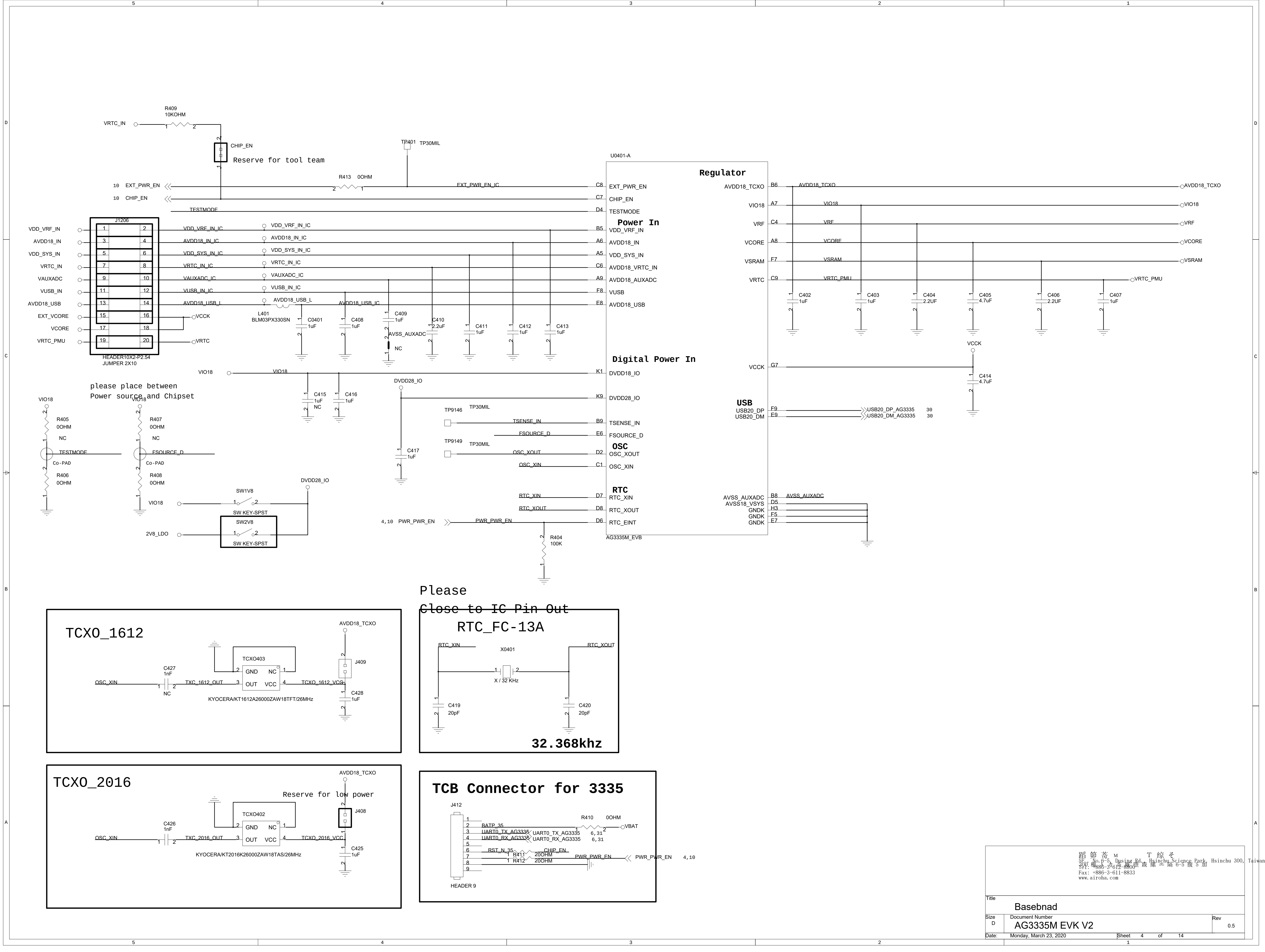
Date: Monday, March 23, 2020

Sheet 2 of 14


External Power
Internal LDO
GPIO Power Source
Connectivity ANT
DUPLEXER
Power IN
AVDD18_IN
AVDD18_USB
VDD_SYS_IN
VDD_VRF_IN
AVDD18_VRTC_IN
AVDD18_AUXADC
VUSB
VCCK
Regulator
AVDD18_TCXO
VIO18
VRF
VCORE
VSRAM
VRTC
Digital PowerIN
DVDD18_IO
DVDD28_IO
RF
RF_IN_L1
LNA_EN_L1
RF_IN_L5
LNA_EN_L5
Frequency
OSC_XIN
OSC_XOUT
RTC
26MHz TCXO
32 kHz Crystal
AG3335M
Interface
I2C_MST0
I2C_MST1
I2C_SLV0
USB2.0
UART_MST0
UART_MST1
UART_MST2
SPI_MST0
SPI_MST1
SPI_SLV0
PWM0
PWM1
PWM2
SDIO2.0
ADC
ADC0
ADC1
ADC2
ADC3
Thermal Sensor
Timing
PPS0
PPS1
UART0
UART1
UART2
nema log
system log
AB1532
SDIO2.0
SDCard
TimePulse
PPS OUT

蹈 笳 蒈 M T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 穝 λ カ 蒈 叇 隉 跋 繧 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title Block Diagrams		
Size D	Document Number AG3335M EVK V2	Rev 0.5
Date: Monday, March 23, 2020	Sheet 3 of 14	

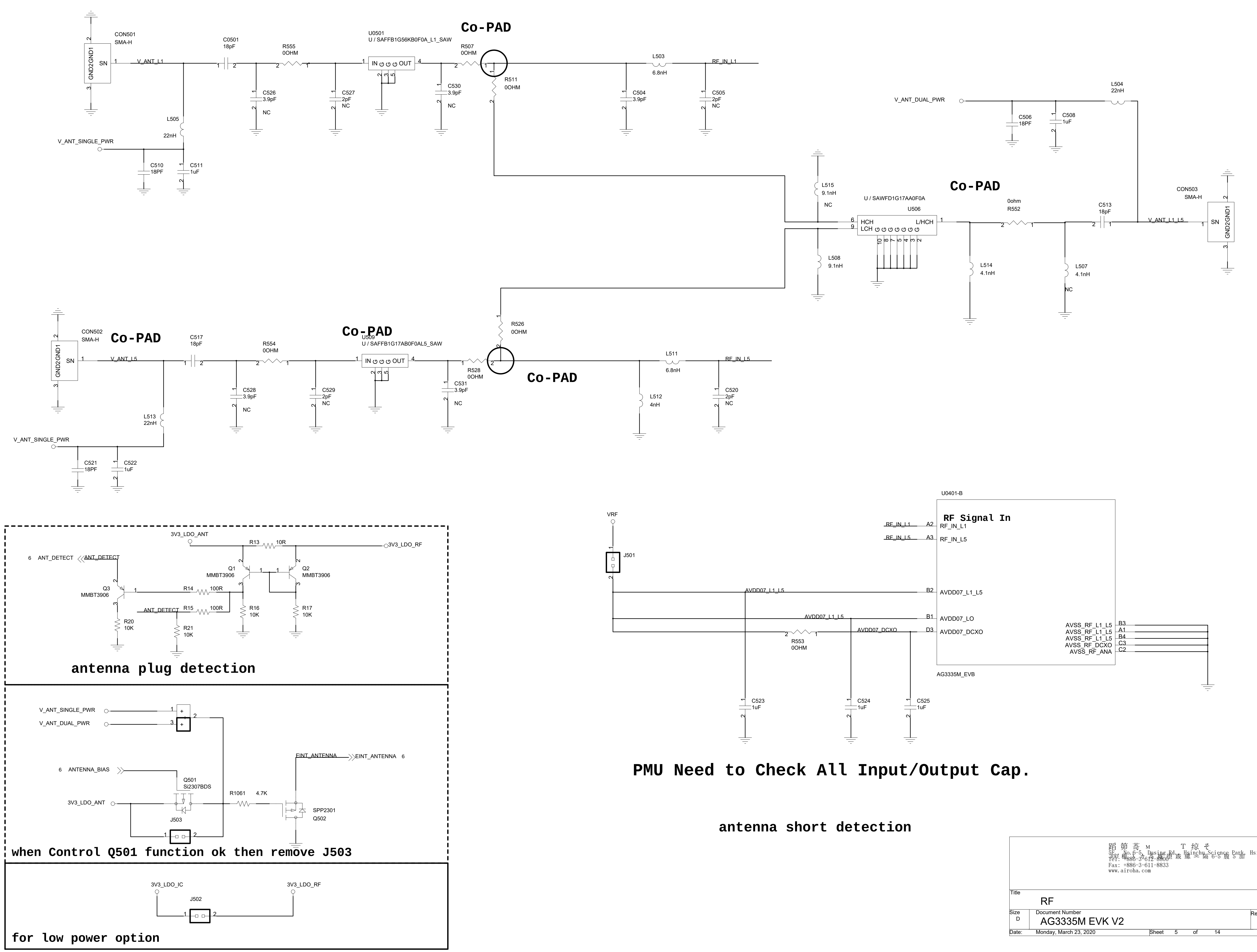

VRTC_IN
R409
10KOHM
CHIP_EN
Reserve for tool team
TP401
TP30MIL
U0401-A
Regulator
EXT_PWR_EN
CHIP_EN
TESTMODE
R413
0OHM
EXT_PWR_EN_IC
J1206
VDD_VRF_IN
AVDD18_IN
VDD_SYS_IN
VRTC_IN
VAUXADC
VUSB_IN
AVDD18_USB
EXT_VCORE
VCORE
VRTC_PMU
HEADER10X2-P2.54
JUMPER 2X10
VDD_VRF_IN_IC
AVDD18_IN_IC
VDD_SYS_IN_IC
VRTC_IN_IC
VAUXADC_IC
VUSB_IN_IC
AVDD18_USB_L
AVDD18_USB_IC
VCCK
VRTC
L401
BLM03PX330SN
C0401 1uF
C408 1uF
C409 1uF
AVSS_AUXADC
NC
C410 2.2uF
C411 1uF
C412 1uF
C413 1uF
Power In
VDD_VRF_IN
AVDD18_IN
VDD_SYS_IN
AVDD18_VRTC_IN
AVDD18_AUXADC
VUSB
AVDD18_USB
AVDD18_TCXO
VIO18
VRF
VCORE
VSRAM
VRTC
VRTC_PMU
C402 1uF
C403 1uF
C404 2.2UF
C405 4.7uF
C406 2.2UF
C407 1uF
VCCK
C414 4.7uF
Digital Power In
DVDD18_IO
DVDD28_IO
VIO18
C415 1uF NC
C416 1uF
DVDD28_IO
C417 1uF
please place between
Power source and Chipset
R405 0OHM NC
R407 0OHM NC
TESTMODE
FSOURCE_D
Co-PAD
R406 0OHM
R408 0OHM
TP9146
TP30MIL
TSENSE_IN
FSOURCE_D
TP9149
TP30MIL
OSC
OSC_XOUT
OSC_XIN
USB
USB20_DP
USB20_DM
USB20_DP_AG3335 30
USB20_DM_AG3335 30
RTC
RTC_XIN
RTC_XOUT
RTC_EINT
AVSS_AUXADC
AVSS18_VSYS
GNDK
AG3335M_EVB
SW1V8
SW KEY-SPST
SW2V8
SW KEY-SPST
VIO18
2V8_LDO
DVDD28_IO
4,10 PWR_PWR_EN
R404 100K
Please
Close to IC Pin Out
TCXO_1612
AVDD18_TCXO
TCXO403
GND NC OUT VCC
C427 1nF NC
OSC_XIN
TXC_1612_OUT
TCXO_1612_VCC
J409
C428 1uF
KYOCERA/KT1612A26000ZAW18TFT/26MHz
RTC_FC-13A
RTC_XIN
RTC_XOUT
X0401
X / 32 KHz
C419 20pF
C420 20pF
32.368khz
TCXO_2016
AVDD18_TCXO
Reserve for low power
TCXO402
J408
C426 1nF
TXC_2016_OUT
TCXO_2016_VCC
C425 1uF
KYOCERA/KT2016K26000ZAW18TAS/26MHz
TCB Connector for 3335
J412
BATP_35
UART0_TX_AG3335
UART0_RX_AG3335
RST_N_35
CHIP_EN
R410 0OHM
VBAT
6,31
R411 20OHM
R412 20OHM
PWR_PWR_EN
4,10
HEADER 9

蹈 節 蕎 M T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 穠 λ カ 蕎 厩 塱 跋 繆 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title	Basebnad	
Size D	Document Number AG3335M EVK V2	Rev 0.5
Date: Monday, March 23, 2020	Sheet 4 of 14	

L0201

[illegible]


CON501
SMA-H
SN
GND2GND1
V_ANT_L1
C0501
18pF
R555
0OHM
U0501
U / SAFFB1G56KB0F0A_L1_SAW
IN G G G OUT
R507
0OHM
R511
0OHM
L503
6.8nH
RF_IN_L1
C526
3.9pF
NC
C527
2pF
NC
C530
3.9pF
NC
C504
3.9pF
C505
2pF
NC
L505
22nH
V_ANT_SINGLE_PWR
C510
18PF
C511
1uF
V_ANT_DUAL_PWR
L504
22nH
C506
18PF
C508
1uF
L515
9.1nH
NC
U / SAWFD1G17AA0F0A
U506
HCH
LCH
L/HCH
0ohm
R552
C513
18pF
V_ANT_L1_L5
CON503
SMA-H
L508
9.1nH
L514
4.1nH
L507
4.1nH
NC
R526
0OHM
CON502
SMA-H
V_ANT_L5
C517
18pF
R554
0OHM
U509
U / SAFFB1G17AB0F0AL5_SAW
R528
0OHM
L511
6.8nH
RF_IN_L5
C528
3.9pF
NC
C529
2pF
NC
C531
3.9pF
NC
L512
4nH
C520
2pF
NC
L513
22nH
C521
18PF
C522
1uF
U0401-B
VRF
J501
RF Signal In
RF_IN_L1
RF_IN_L5
AVDD07_L1_L5
AVDD07_LO
AVDD07_DCXO
AVSS_RF_L1_L5
AVSS_RF_DCXO
AVSS_RF_ANA
A2
A3
B2
B1
D3
B3
A1
B4
C3
C2
AG3335M_EVB
R553
0OHM
C523
1uF
C524
1uF
C525
1uF
3V3_LDO_ANT
R13
10R
3V3_LDO_RF
6 ANT_DETECT
ANT_DETECT
Q1
MMBT3906
Q2
MMBT3906
Q3
MMBT3906
R14
100R
R15
100R
R16
10K
R17
10K
R20
10K
R21
10K
V_ANT_SINGLE_PWR
V_ANT_DUAL_PWR
6 ANTENNA_BIAS
Q501
Si2307BDS
3V3_LDO_ANT
J503
R1061
4.7K
EINT_ANTENNA
EINT_ANTENNA 6
SPP2301
Q502
3V3_LDO_IC
3V3_LDO_RF
J502
L0603SMD/0402
C0201

Co-PAD

Co-PAD

Co-PAD

Co-PAD

Co-PAD

antenna plug detection

when Control Q501 function ok then remove J503

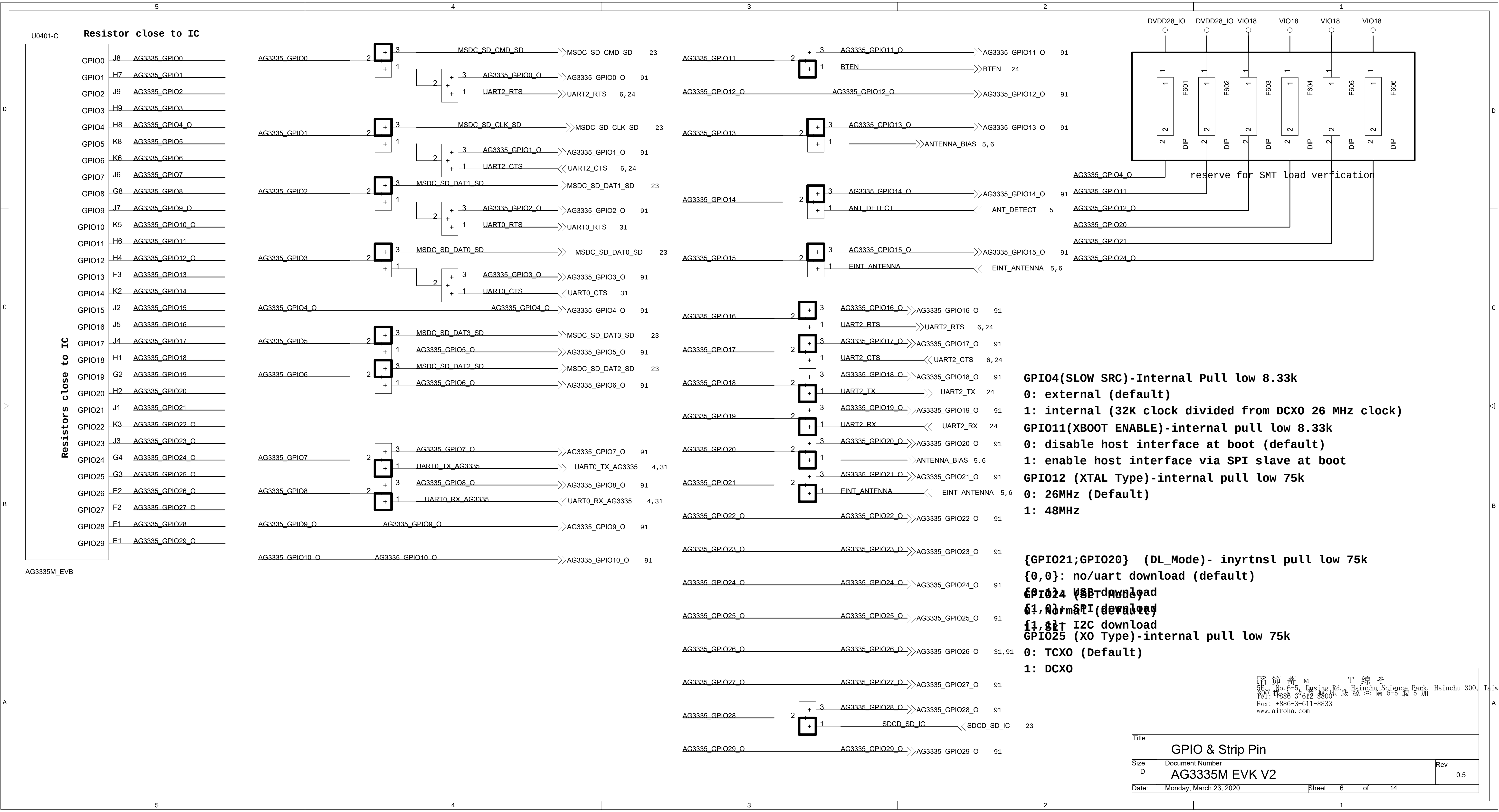
for low power option

PMU Need to Check All Input/Output Cap.

antenna short detection

蹈 篩 蒿 M　T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 穩 λ カ 蒿 蔵 隍 跋 縲 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title: RF
Size: D
Document Number: AG3335M EVK V2
Rev: 0.5
Date: Monday, March 23, 2020
Sheet 5 of 14


U0401-C
Resistor close to IC
Resistors close to IC
AG3335M_EVB
GPIO0 J8 AG3335_GPIO0
GPIO1 H7 AG3335_GPIO1
GPIO2 J9 AG3335_GPIO2
GPIO3 H9 AG3335_GPIO3
GPIO4 H8 AG3335_GPIO4_O
GPIO5 K8 AG3335_GPIO5
GPIO6 K6 AG3335_GPIO6
GPIO7 J6 AG3335_GPIO7
GPIO8 G8 AG3335_GPIO8
GPIO9 J7 AG3335_GPIO9_O
GPIO10 K5 AG3335_GPIO10_O
GPIO11 H6 AG3335_GPIO11
GPIO12 H4 AG3335_GPIO12_O
GPIO13 F3 AG3335_GPIO13
GPIO14 K2 AG3335_GPIO14
GPIO15 J2 AG3335_GPIO15
GPIO16 J5 AG3335_GPIO16
GPIO17 J4 AG3335_GPIO17
GPIO18 H1 AG3335_GPIO18
GPIO19 G2 AG3335_GPIO19
GPIO20 H2 AG3335_GPIO20
GPIO21 J1 AG3335_GPIO21
GPIO22 K3 AG3335_GPIO22_O
GPIO23 J3 AG3335_GPIO23_O
GPIO24 G4 AG3335_GPIO24_O
GPIO25 G3 AG3335_GPIO25_O
GPIO26 E2 AG3335_GPIO26_O
GPIO27 F2 AG3335_GPIO27_O
GPIO28 F1 AG3335_GPIO28
GPIO29 E1 AG3335_GPIO29_O
MSDC_SD_CMD_SD 23
AG3335_GPIO0_O 91
UART2_RTS 6,24
MSDC_SD_CLK_SD 23
AG3335_GPIO1_O 91
UART2_CTS 6,24
MSDC_SD_DAT1_SD 23
AG3335_GPIO2_O 91
UART0_RTS 31
MSDC_SD_DAT0_SD 23
AG3335_GPIO3_O 91
UART0_CTS 31
AG3335_GPIO4_O 91
MSDC_SD_DAT3_SD 23
AG3335_GPIO5_O 91
MSDC_SD_DAT2_SD 23
AG3335_GPIO6_O 91
AG3335_GPIO7_O 91
UART0_TX_AG3335 4,31
AG3335_GPIO8_O 91
UART0_RX_AG3335 4,31
AG3335_GPIO9_O 91
AG3335_GPIO10_O 91
AG3335_GPIO11_O 91
BTEN 24
AG3335_GPIO12_O 91
AG3335_GPIO13_O 91
ANTENNA_BIAS 5,6
AG3335_GPIO14_O 91
ANT_DETECT 5
AG3335_GPIO15_O 91
EINT_ANTENNA 5,6
AG3335_GPIO16_O 91
UART2_RTS 6,24
AG3335_GPIO17_O 91
UART2_CTS 6,24
AG3335_GPIO18_O 91
UART2_TX 24
AG3335_GPIO19_O 91
UART2_RX 24
AG3335_GPIO20_O 91
ANTENNA_BIAS 5,6
AG3335_GPIO21_O 91
EINT_ANTENNA 5,6
AG3335_GPIO22_O 91
AG3335_GPIO23_O 91
AG3335_GPIO24_O 91
AG3335_GPIO25_O 91
AG3335_GPIO26_O 31,91
AG3335_GPIO27_O 91
AG3335_GPIO28_O 91
SDCD_SD_IC 23
AG3335_GPIO29_O 91
DVDD28_IO
DVDD28_IO
VIO18
VIO18
VIO18
VIO18
F601 DIP
F602 DIP
F603 DIP
F604 DIP
F605 DIP
F606 DIP
AG3335_GPIO4_O
AG3335_GPIO11
AG3335_GPIO12_O
AG3335_GPIO20
AG3335_GPIO21
AG3335_GPIO24_O
reserve for SMT load verfication

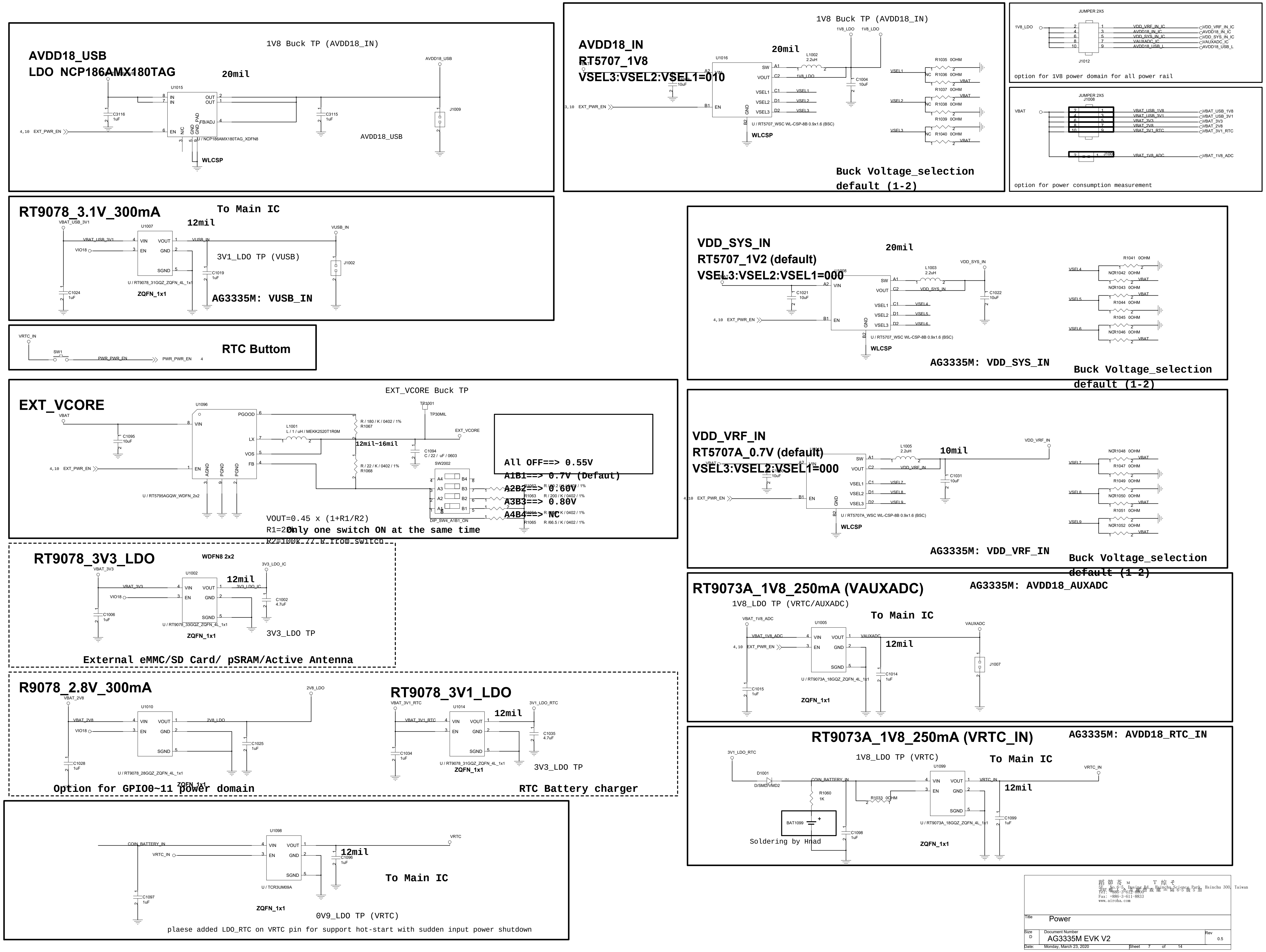
GPIO4(SLOW SRC)-Internal Pull low 8.33k
0: external (default)
1: internal (32K clock divided from DCXO 26 MHz clock)
GPIO11(XBOOT ENABLE)-internal pull low 8.33k
0: disable host interface at boot (default)
1: enable host interface via SPI slave at boot
GPIO12 (XTAL Type)-internal pull low 75k
0: 26MHz (Default)
1: 48MHz

{GPIO21;GPIO20} (DL_Mode)- inyrtnsl pull low 75k
{0,0}: no/uart download (default)
{0,1}: USB download
{1,0}: SPI download
{1,1}: I2C download

GPIO24 (SET Mode)
0: Normal (default)
1: SET
GPIO25 (XO Type)-internal pull low 75k
0: TCXO (Default)
1: DCXO

踞 節 莕 M T 綜 そ
5F, No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title: GPIO & Strip Pin
Size: D
Document Number: AG3335M EVK V2
Rev: 0.5
Date: Monday, March 23, 2020
Sheet 6 of 14

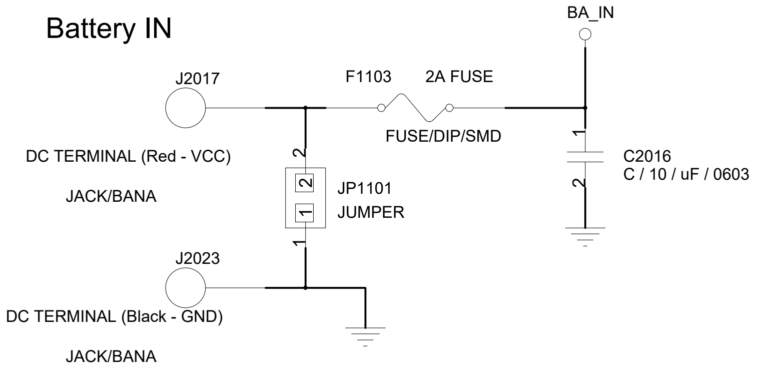

AVDD18_USB
LDO NCP186AMX180TAG
1V8 Buck TP (AVDD18_IN)
20mil
U1015
U / NCP186AMX180TAG_XDFN8
WLCSP
AVDD18_USB
J1009
4,10 EXT_PWR_EN
C3116 1uF
C3115 1uF
AVDD18_IN
RT5707_1V8
VSEL3:VSEL2:VSEL1=010
1V8 Buck TP (AVDD18_IN)
U1016
L1002 2.2uH
1V8_LDO
C1004 10uF
U / RT5707_WSC WL-CSP-8B 0.9x1.6 (BSC)
WLCSP
Buck Voltage_selection
default (1-2)
R1035 0OHM
R1036 0OHM
R1037 0OHM
R1038 0OHM
R1039 0OHM
R1040 0OHM
VSEL1
VSEL2
VSEL3
VBAT
JUMPER 2X5
J1012
1V8_LDO
VDD_VRF_IN_IC
AVDD18_IN_IC
VDD_SYS_IN_IC
VAUXADC_IC
AVDD18_USB_L
option for 1V8 power domain for all power rail
JUMPER 2X5
J1008
VBAT_USB_1V8
VBAT_USB_3V1
VBAT_3V3
VBAT_2V8
VBAT_3V1_RTC
VBAT_1V8_ADC
J1006
option for power consumption measurement
RT9078_3.1V_300mA
To Main IC
12mil
VBAT_USB_3V1
U1007
VUSB_IN
VIO18
3V1_LDO TP (VUSB)
C1019 1uF
C1024 1uF
J1002
U / RT9078_31GQZ_ZQFN_4L_1x1
ZQFN_1x1
AG3335M: VUSB_IN
VRTC_IN
SW1
PWR_PWR_EN
RTC Buttom
EXT_VCORE
EXT_VCORE Buck TP
U1096
VBAT
C1095 10uF
L1001
L / 1 / uH / MEKK2520T1R0M
R / 180 / K / 0402 / 1%
R1067
R / 22 / K / 0402 / 1%
R1068
TP1001
TP30MIL
EXT_VCORE
12mil~16mil
C1094
C / 22 / uF / 0603
SW2002
DIP_SW4_A1B1_ON
R1062
R1063
R1064
R1065
U / RT5795AGQW_WDFN_2x2
All OFF==> 0.55V
A1B1==> 0.7V (Defaut)
A2B2==> 0.60V
A3B3==> 0.80V
A4B4==> NC
VOUT=0.45 x (1+R1/R2)
R1=200k
R2=100k // R from switch
Only one switch ON at the same time
VDD_SYS_IN
RT5707_1V2 (default)
VSEL3:VSEL2:VSEL1=000
20mil
U1008
L1003 2.2uH
C1021 10uF
C1022 10uF
U / RT5707_WSC WL-CSP-8B 0.9x1.6 (BSC)
WLCSP
AG3335M: VDD_SYS_IN
Buck Voltage_selection
default (1-2)
R1041 0OHM
R1042 0OHM
R1043 0OHM
R1044 0OHM
R1045 0OHM
R1046 0OHM
VSEL4
VSEL5
VSEL6
VDD_VRF_IN
RT5707A_0.7V (default)
VSEL3:VSEL2:VSEL1=000
10mil
L1005 2.2uH
C1031 10uF
U / RT5707A_WSC WL-CSP-8B 0.9x1.6 (BSC)
WLCSP
AG3335M: VDD_VRF_IN
Buck Voltage_selection
default (1-2)
R1047 0OHM
R1048 0OHM
R1049 0OHM
R1050 0OHM
R1051 0OHM
R1052 0OHM
VSEL7
VSEL8
VSEL9
RT9078_3V3_LDO
WDFN8 2x2
VBAT_3V3
U1002
12mil
3V3_LDO_IC
C1002 4.7uF
C1006 1uF
U / RT9078_33GQZ_ZQFN_4L_1x1
ZQFN_1x1
3V3_LDO TP
External eMMC/SD Card/ pSRAM/Active Antenna
RT9073A_1V8_250mA (VAUXADC)
AG3335M: AVDD18_AUXADC
1V8_LDO TP (VRTC/AUXADC)
To Main IC
VBAT_1V8_ADC
U1005
VAUXADC
12mil
J1007
C1014 1uF
C1015 1uF
U / RT9073A_18GQZ_ZQFN_4L_1x1
ZQFN_1x1
R9078_2.8V_300mA
VBAT_2V8
U1010
2V8_LDO
C1025 1uF
C1028 1uF
U / RT9078_28GQZ_ZQFN_4L_1x1
ZQFN_1x1
Option for GPIO0~11 power domain
RT9078_3V1_LDO
VBAT_3V1_RTC
U1014
12mil
3V1_LDO_RTC
C1035 4.7uF
C1034 1uF
U / RT9078_31GQZ_ZQFN_4L_1x1
ZQFN_1x1
3V3_LDO TP
RTC Battery charger
RT9073A_1V8_250mA (VRTC_IN)
AG3335M: AVDD18_RTC_IN
1V8_LDO TP (VRTC)
To Main IC
3V1_LDO_RTC
D1001
D/SMD/VMD2
COIN_BATTERY_IN
R1060 1K
R1033 0OHM
BAT1099
Soldering by Hnad
C1098 1uF
U1099
VRTC_IN
12mil
C1099 1uF
U / RT9073A_18GQZ_ZQFN_4L_1x1
ZQFN_1x1
U1098
COIN_BATTERY_IN
VRTC_IN
VRTC
12mil
C1096 1uF
C1097 1uF
U / TCR3UM09A
ZQFN_1x1
To Main IC
0V9_LDO TP (VRTC)
plaese added LDO_RTC on VRTC pin for support hot-start with sudden input power shutdown

蹈 節 蒿 M T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 梯 λ 力 斉 興 陧 跛 縲 ≪ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

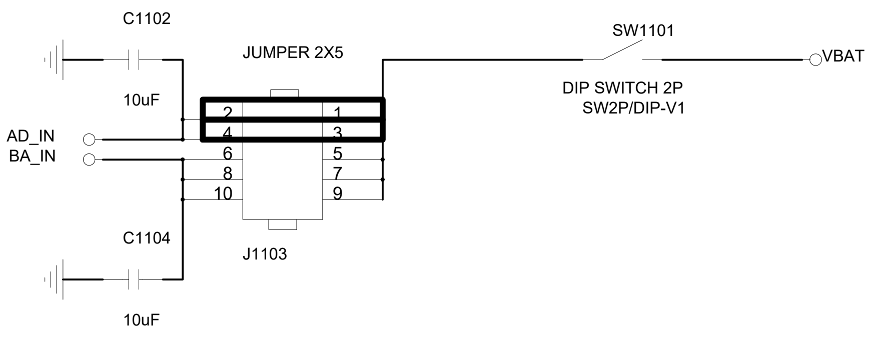
Title	Power		
Size D	Document Number AG3335M EVK V2		Rev 0.5
Date:	Monday, March 23, 2020	Sheet 7 of 14	

R0402/SMD
C0.22F3.3V/SMD/EN-RL
D/SMD/VMD2

DC POWER IN from BATTERY


Battery IN
BA_IN
J2017
F1103
2A FUSE
FUSE/DIP/SMD
DC TERMINAL (Red - VCC)
JACK/BANA
2
1
JP1101
JUMPER
C2016
C / 10 / uF / 0603
1
2
J2023
DC TERMINAL (Black - GND)
JACK/BANA

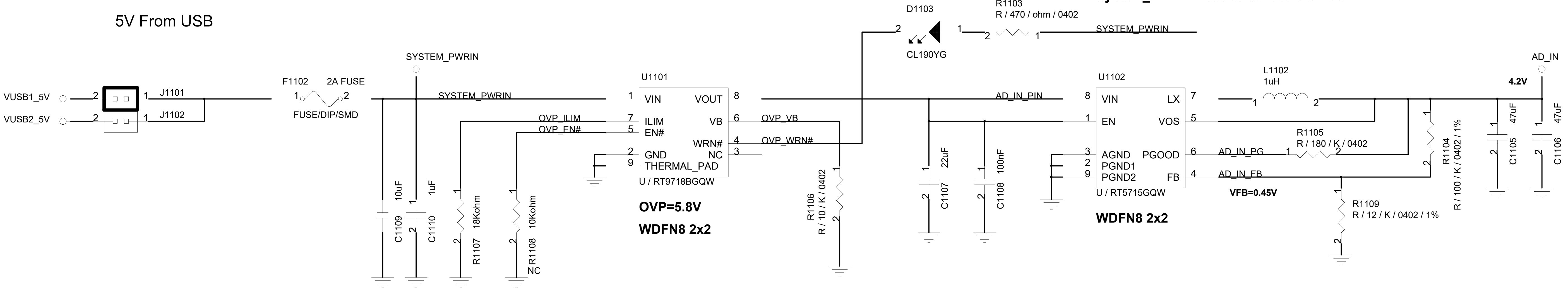
power optipn


C1102
10uF
JUMPER 2X5
SW1101
VBAT
DIP SWITCH 2P
SW2P/DIP-V1
AD_IN
BA_IN
2
1
4
3
6
5
8
7
10
9
J1103
C1104
10uF

VBAT/AD_IN/BA_IN Power Plan need 120mm

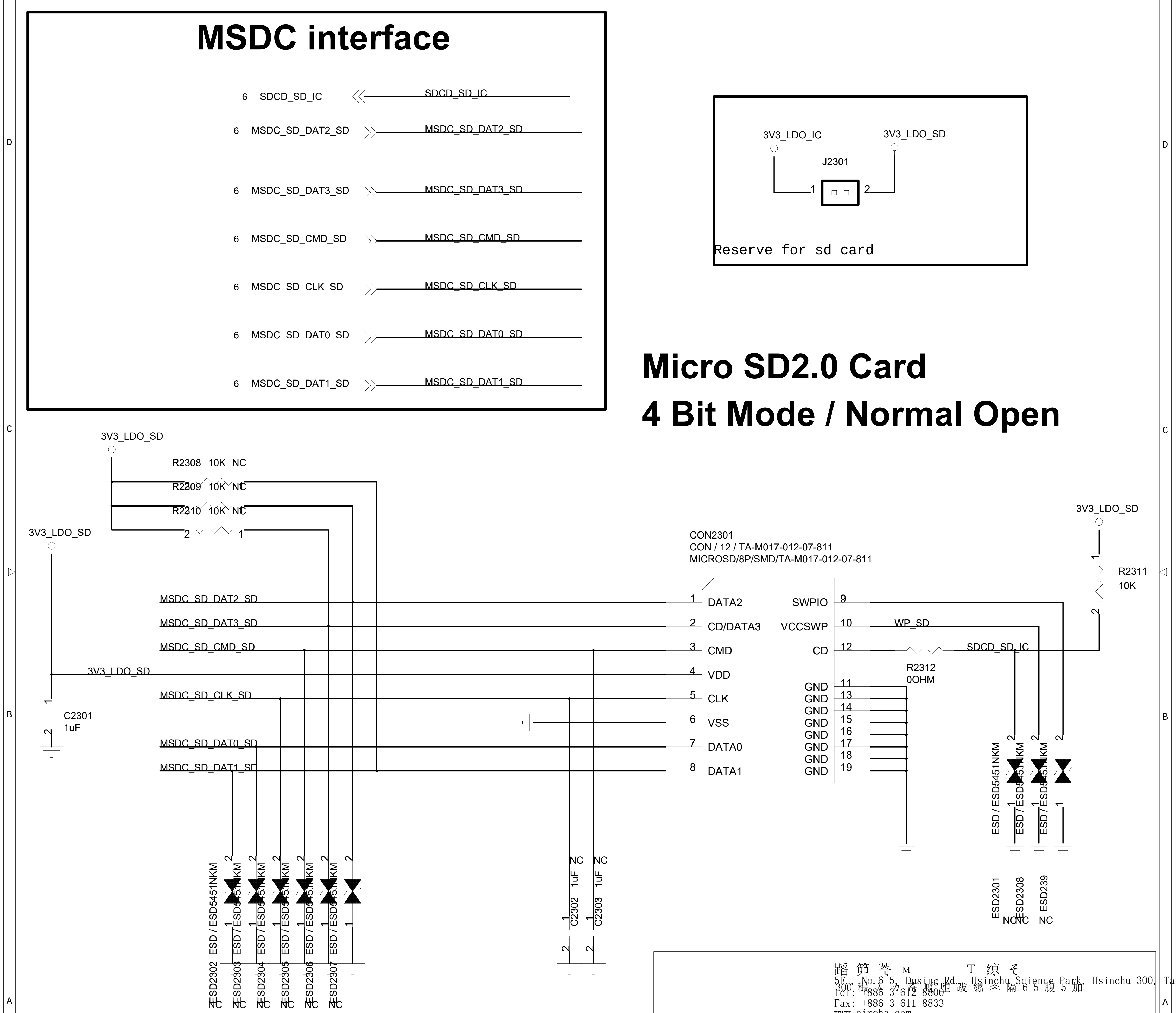
DC POWER IN From Adaptor or USB (AD_IN)

System_PWRIN need to be less than 5.8V


5V From USB
VUSB1_5V
VUSB2_5V
J1101
J1102
F1102
2A FUSE
FUSE/DIP/SMD
SYSTEM_PWRIN
SYSTEM_PWRIN
C1109
10uF
C1110
1uF
R1107
18Kohm
R1108
10Kohm
NC
OVP_ILIM
OVP_EN#
U1101
VIN
VOUT
ILIM
EN#
VB
WRN#
GND
NC
THERMAL_PAD
U / RT9718BGQW
OVP_VB
OVP_WRN#
R1106
R / 10 / K / 0402
OVP=5.8V
WDFN8 2x2
D1103
CL190YG
R1103
R / 470 / ohm / 0402
SYSTEM_PWRIN
AD_IN_PIN
C1107
22uF
C1108
100nF
U1102
VIN
LX
EN
VOS
AGND
PGND1
PGND2
PGOOD
FB
U / RT5715GQW
WDFN8 2x2
L1102
1uH
R1105
R / 180 / K / 0402
AD_IN_PG
AD_IN_FB
VFB=0.45V
R1109
R / 12 / K / 0402 / 1%
R1104
R / 100 / K / 0402 / 1%
AD_IN
4.2V
C1105
47uF
C1106
47uF

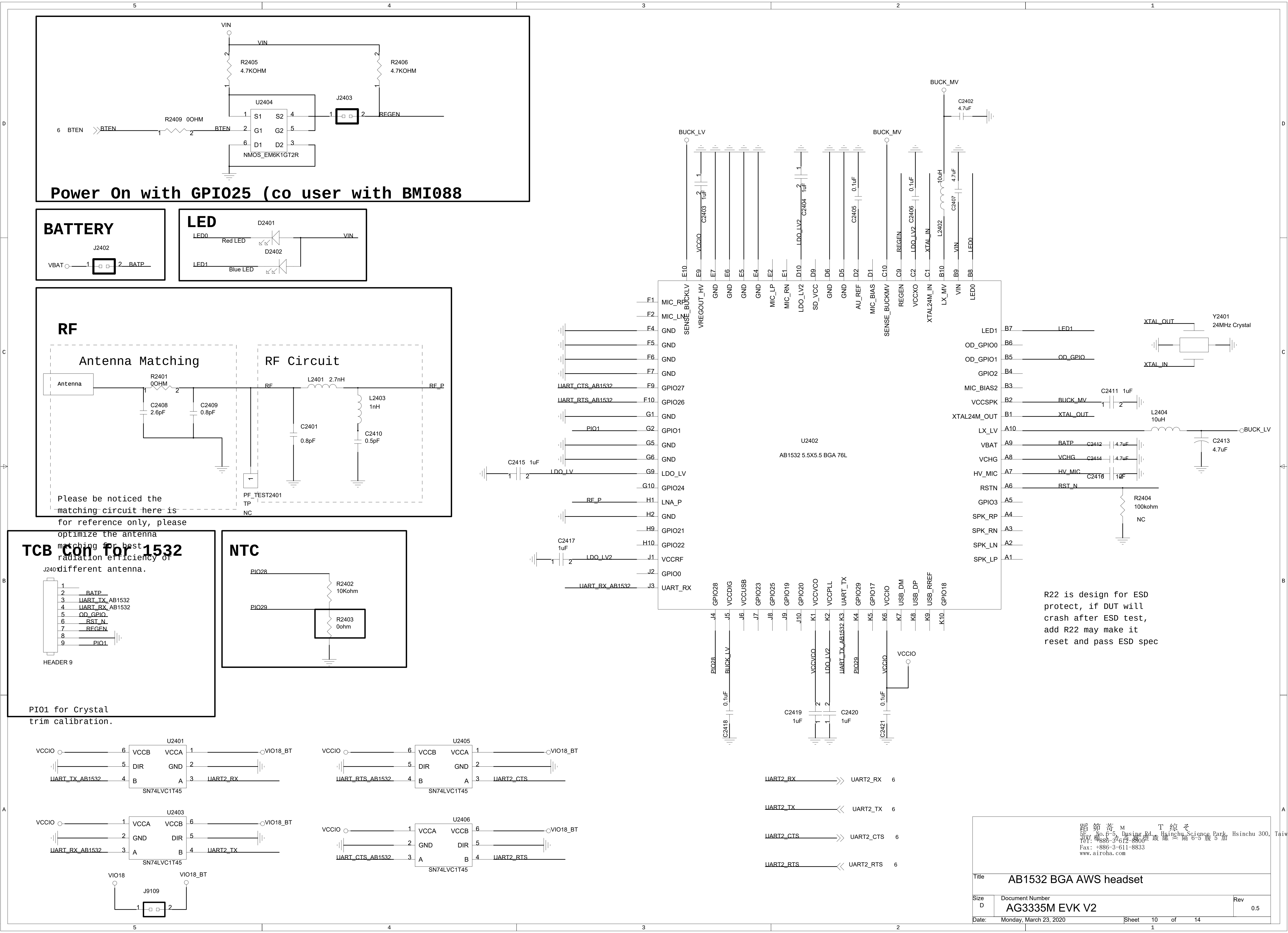
蹈 篛 蒥 M T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 穛 λ カ 蒥 厩 隍 跋 繶 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title Power		
Size D	Document Number AG3335M EVK V2	Rev 0.0
Date:	Monday, March 23, 2020	Sheet 8 of 14


MSDC interface
6 SDCD_SD_IC
SDCD_SD_IC
6 MSDC_SD_DAT2_SD
MSDC_SD_DAT2_SD
6 MSDC_SD_DAT3_SD
MSDC_SD_DAT3_SD
6 MSDC_SD_CMD_SD
MSDC_SD_CMD_SD
6 MSDC_SD_CLK_SD
MSDC_SD_CLK_SD
6 MSDC_SD_DAT0_SD
MSDC_SD_DAT0_SD
6 MSDC_SD_DAT1_SD
MSDC_SD_DAT1_SD
3V3_LDO_IC
3V3_LDO_SD
J2301
1
2
Reserve for sd card
Micro SD2.0 Card
4 Bit Mode / Normal Open
3V3_LDO_SD
R2308 10K NC
R2309 10K NC
R2310 10K NC
2
1
3V3_LDO_SD
CON2301
CON / 12 / TA-M017-012-07-811
MICROSD/8P/SMD/TA-M017-012-07-811
MSDC_SD_DAT2_SD
MSDC_SD_DAT3_SD
MSDC_SD_CMD_SD
3V3_LDO_SD
MSDC_SD_CLK_SD
MSDC_SD_DAT0_SD
MSDC_SD_DAT1_SD
C2301
1uF
1 DATA2
2 CD/DATA3
3 CMD
4 VDD
5 CLK
6 VSS
7 DATA0
8 DATA1
SWPIO 9
VCCSWP 10
CD 12
GND 11
GND 13
GND 14
GND 15
GND 16
GND 17
GND 18
GND 19
WP_SD
SDCD_SD_IC
R2312
0OHM
3V3_LDO_SD
R2311
10K
C2302 1uF NC
C2303 1uF NC
ESD / ESD5451NKM
ESD2302 NC
ESD2303 NC
ESD2304 NC
ESD2305 NC
ESD2306 NC
ESD2307 NC
ESD2301 NC
ESD2308 NC
ESD239 NC
5
4
3
2
1
A
B
C
D

蹈 笷 蒿 M T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 橇 λ カ 蒿 厩 隍 跋 繆 ≪ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title	SD Card	
Size D	Document Number AG3335M EVK V2	Rev 0.5
Date:	Monday, March 23, 2020	Sheet 9 of 14


Power On with GPIO25 (co user with BMI088
VIN
R2405
4.7KOHM
R2406
4.7KOHM
U2404
S1
S2
G1
G2
D1
D2
NMOS_EM6K1GT2R
J2403
REGEN
6 BTEN
BTEN
R2409 0OHM
BATTERY
J2402
VBAT
BATP
LED
D2401
LED0
Red LED
D2402
LED1
Blue LED
RF
Antenna Matching
RF Circuit
Antenna
R2401
0OHM
C2408
2.6pF
C2409
0.8pF
RF
L2401 2.7nH
RF_P
L2403
1nH
C2401
0.8pF
C2410
0.5pF
PF_TEST2401
TP
NC
TCB Con for 1532
J2401
BATP
UART_TX_AB1532
UART_RX_AB1532
OD_GPIO
RST_N
REGEN
PIO1
HEADER 9
NTC
PIO28
R2402
10Kohm
PIO29
R2403
0ohm
BUCK_LV
BUCK_MV
C2402
4.7uF
C2403
1uF
C2404
1uF
C2405
0.1uF
C2406
0.1uF
L2402
10uH
C2407
4.7uF
VCCIO
LDO_LV2
REGEN
XTAL_IN
VIN
LED0
E10 E9 E7 E6 E5 E4 E2 E1 D10 D9 D6 D5 D2 D1 C10 C9 C2 C1 B10 B9 B8
SENSE_BUCKLV
VREGOUT_HV
GND
MIC_LP
MIC_RN
LDO_LV2
SD_VCC
AU_REF
MIC_BIAS
SENSE_BUCKMV
VCCXO
XTAL24M_IN
LX_MV
F1 MIC_RP
F2 MIC_LN
F4 GND
F5 GND
F6 GND
F7 GND
F9 GPIO27 UART_CTS_AB1532
F10 GPIO26 UART_RTS_AB1532
G1 GND
G2 GPIO1 PIO1
G5 GND
G6 GND
C2415 1uF
G9 LDO_LV
G10 GPIO24
H1 LNA_P RF_P
H2 GND
H9 GPIO21
H10 GPIO22
C2417
1uF
J1 VCCRF LDO_LV2
J2 GPIO0
J3 UART_RX UART_RX_AB1532
U2402
AB1532 5.5X5.5 BGA 76L
LED1 B7 LED1
OD_GPIO0 B6
OD_GPIO1 B5 OD_GPIO
GPIO2 B4
MIC_BIAS2 B3
VCCSPK B2 BUCK_MV
C2411 1uF
XTAL24M_OUT B1 XTAL_OUT
LX_LV A10
L2404
10uH
C2413
4.7uF
BUCK_LV
VBAT A9 BATP C2412 4.7uF
VCHG A8 VCHG C2414 4.7uF
HV_MIC A7 HV_MIC C2416 1uF
RSTN A6 RST_N
R2404
100kohm
NC
GPIO3 A5
SPK_RP A4
SPK_RN A3
SPK_LN A2
SPK_LP A1
Y2401
24MHz Crystal
XTAL_OUT
XTAL_IN
GPIO28 VCCDIG VCCUSB GPIO23 GPIO25 GPIO19 GPIO20 VCCVCO VCCPLL UART_TX GPIO29 GPIO17 VCCIO USB_DM USB_DP USB_RREF GPIO18
J4 J5 J6 J7 J8 J9 J10 K1 K2 K3 K4 K5 K6 K7 K8 K9 K10
PIO28
BUCK_LV
VCCVCO
LDO_LV2
UART_TX_AB1532
PIO29
VCCIO
C2418
0.1uF
C2419
1uF
C2420
1uF
C2421
0.1uF
U2401
VCCIO
VCCB
VCCA
DIR
GND
B
A
VIO18_BT
UART_TX_AB1532
UART2_RX
SN74LVC1T45
U2405
UART_RTS_AB1532
UART2_CTS
U2403
UART_RX_AB1532
UART2_TX
U2406
UART_CTS_AB1532
UART2_RTS
UART2_RX 6
UART2_TX 6
UART2_CTS 6
UART2_RTS 6
VIO18
J9109

Please be noticed the matching circuit here is for reference only, please optimize the antenna matching for best radiation efficiency of different antenna.

PIO1 for Crystal trim calibration.

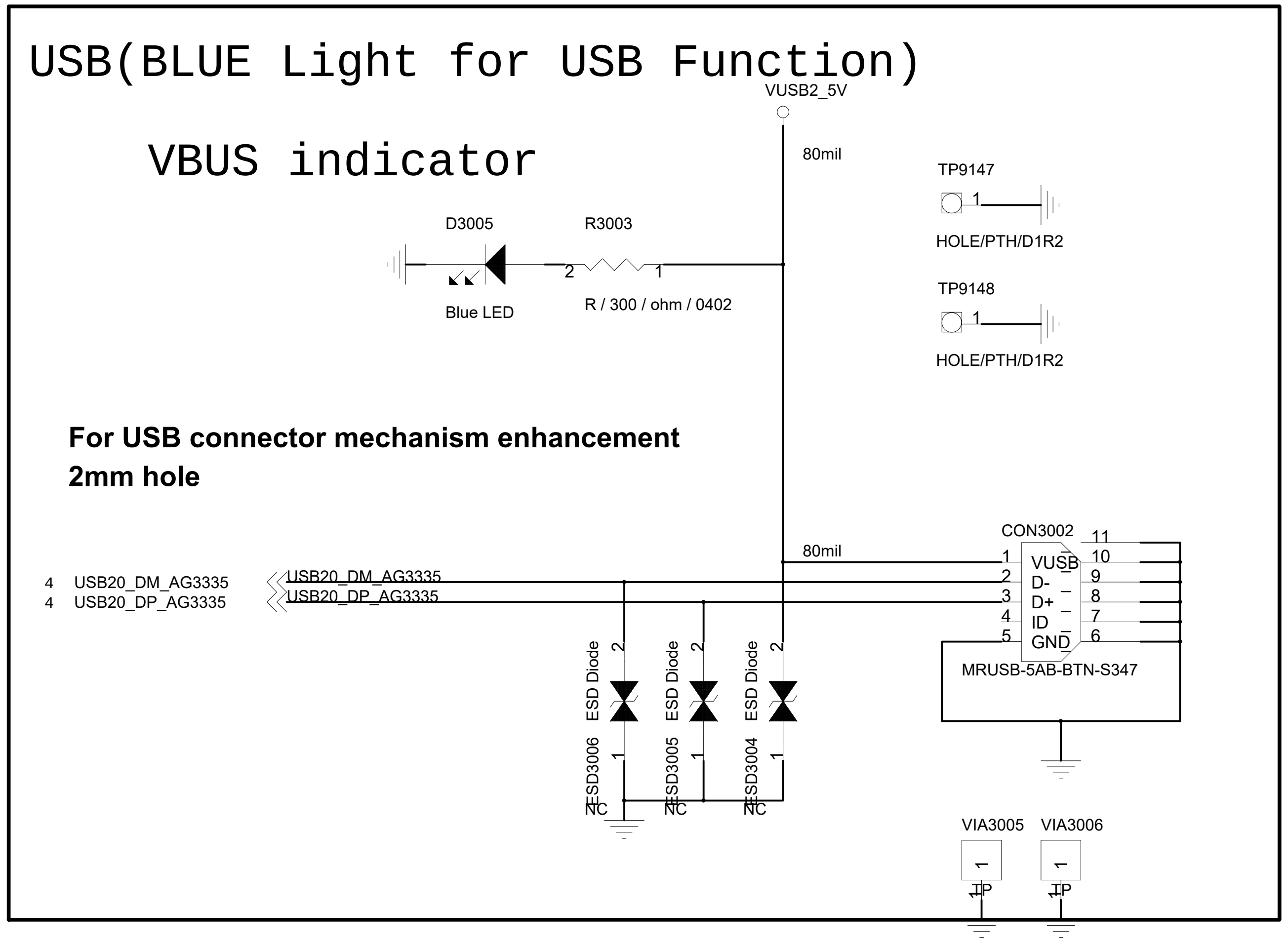
R22 is design for ESD protect, if DUT will crash after ESD test, add R22 may make it reset and pass ESD spec

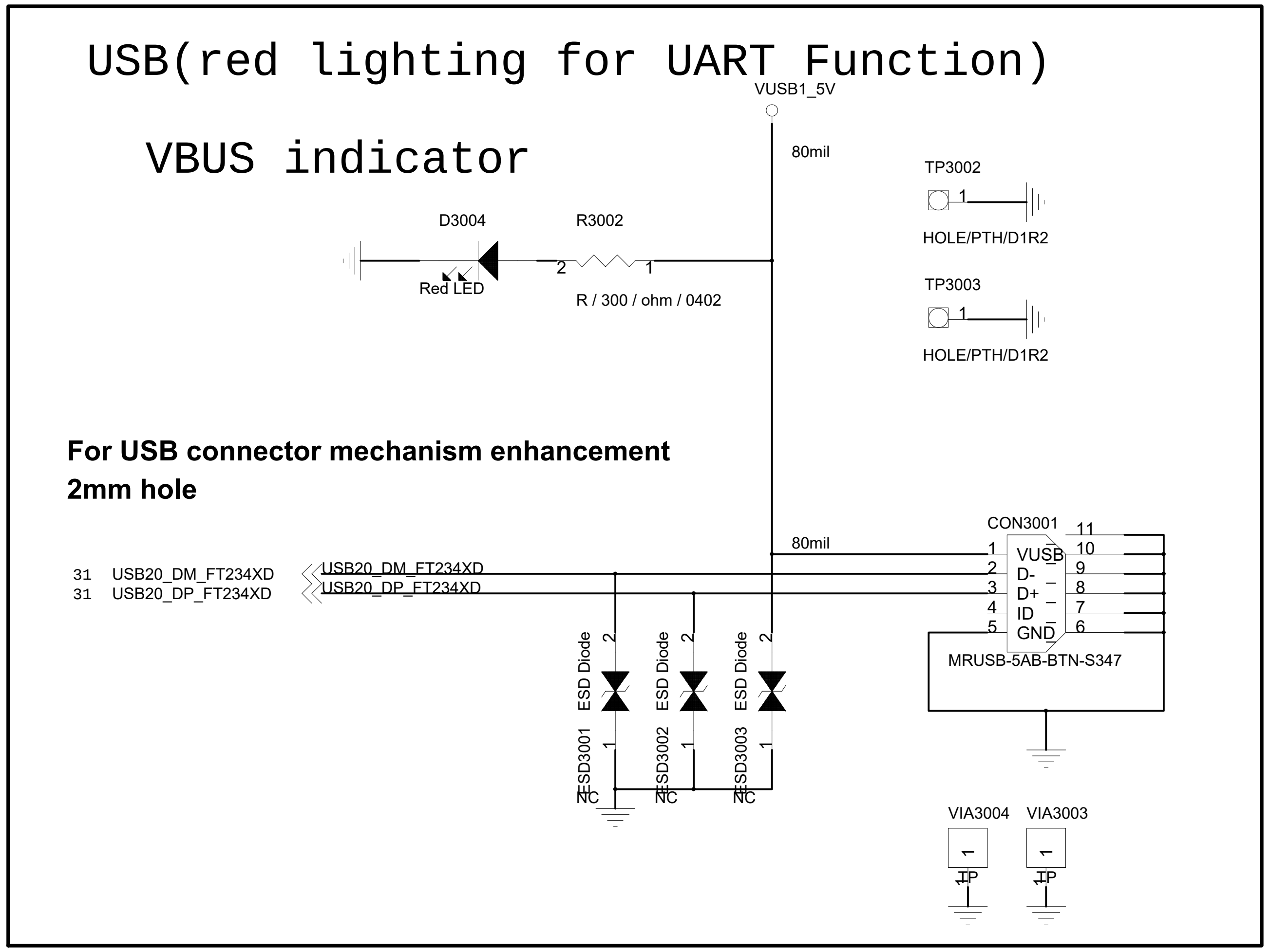
蹈 節 蒚 M T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 穣 λ カ 蒚 厫 隍 跛 縲 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title AB1532 BGA AWS headset

Size D
Document Number AG3335M EVK V2
Rev 0.5

Date: Monday, March 23, 2020
Sheet 10 of 14


USB(BLUE Light for USB Function)
VBUS indicator
VUSB2_5V
80mil
D3005
R3003
2
1
Blue LED
R / 300 / ohm / 0402
TP9147
1
HOLE/PTH/D1R2
TP9148
1
HOLE/PTH/D1R2
For USB connector mechanism enhancement
2mm hole
4 USB20_DM_AG3335
4 USB20_DP_AG3335
USB20_DM_AG3335
USB20_DP_AG3335
80mil
CON3002
1 VUSB
2 D-
3 D+
4 ID
5 GND
11
10
9
8
7
6
MRUSB-5AB-BTN-S347
ESD Diode
ESD3006
ESD3005
ESD3004
NC
VIA3005
VIA3006
TP


USB(red lighting for UART Function)
VBUS indicator
VUSB1_5V
80mil
D3004
R3002
2
1
Red LED
R / 300 / ohm / 0402
TP3002
1
HOLE/PTH/D1R2
TP3003
1
HOLE/PTH/D1R2
For USB connector mechanism enhancement
2mm hole
31 USB20_DM_FT234XD
31 USB20_DP_FT234XD
USB20_DM_FT234XD
USB20_DP_FT234XD
80mil
CON3001
1 VUSB
2 D-
3 D+
4 ID
5 GND
11
10
9
8
7
6
MRUSB-5AB-BTN-S347
ESD Diode
ESD3001
ESD3002
ESD3003
NC
VIA3004
VIA3003
TP

蹈 節 蒿 M T 綜 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 樞 ⅄ カ 蒿 厩 埴 跋 縲 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title USB

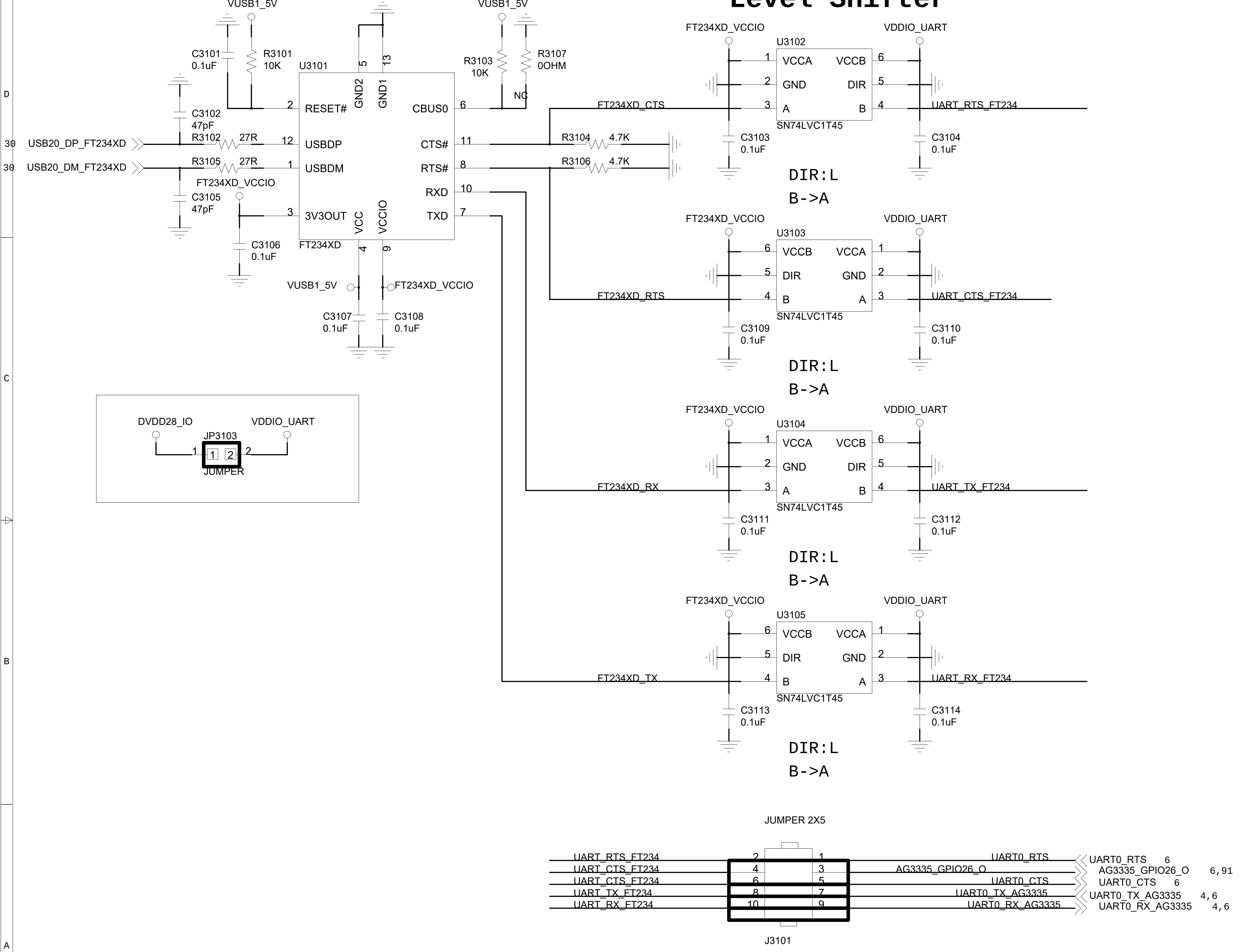
Size D

Document Number AG3335M EVK V2

Rev 0.5

Date: Monday, March 23, 2020 Sheet 11 of 14

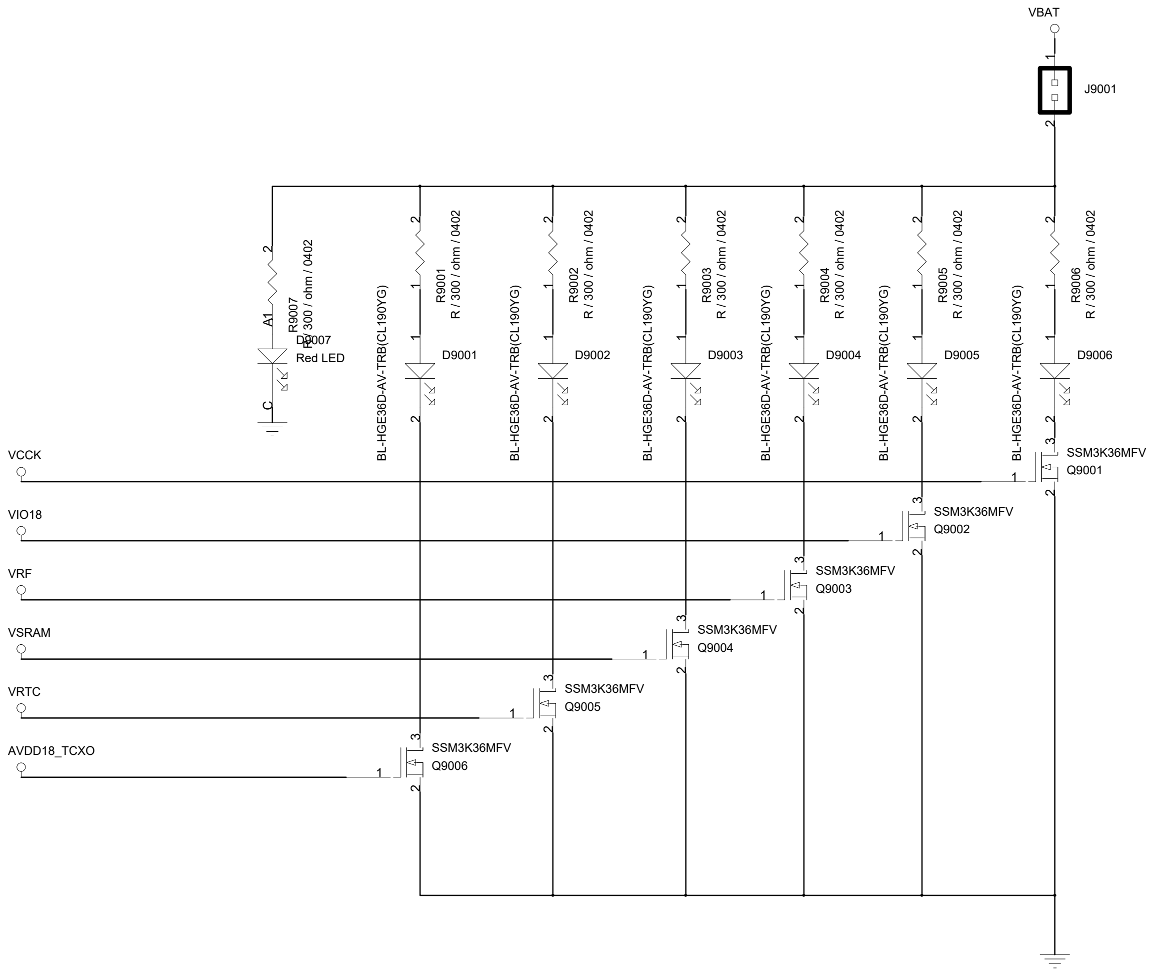
Level Shifter


VUSB1_5V
C3101
0.1uF
R3101
10K
U3101
RESET#
CBUS0
GND2
GND1
5
13
2
6
VUSB1_5V
R3103
10K
R3107
0OHM
NC
C3102
47pF
30 USB20_DP_FT234XD
30 USB20_DM_FT234XD
R3102 27R
R3105 27R
12 USBDP
1 USBDM
CTS# 11
RTS# 8
RXD 10
TXD 7
FT234XD_VCCIO
C3105
47pF
3 3V3OUT
VCC
VCCIO
FT234XD
4
9
C3106
0.1uF
VUSB1_5V
FT234XD_VCCIO
C3107
0.1uF
C3108
0.1uF
R3104 4.7K
R3106 4.7K
FT234XD_CTS
FT234XD_RTS
FT234XD_RX
FT234XD_TX
FT234XD_VCCIO
VDDIO_UART
U3102
VCCA
VCCB
GND
DIR
A
B
SN74LVC1T45
C3103
0.1uF
C3104
0.1uF
UART_RTS_FT234
DIR:L
B->A
U3103
C3109
0.1uF
C3110
0.1uF
UART_CTS_FT234
U3104
C3111
0.1uF
C3112
0.1uF
UART_TX_FT234
U3105
C3113
0.1uF
C3114
0.1uF
UART_RX_FT234
DVDD28_IO
VDDIO_UART
JP3103
JUMPER
JUMPER 2X5
UART_RTS_FT234
UART_CTS_FT234
UART_CTS_FT234
UART_TX_FT234
UART_RX_FT234
UART0_RTS
AG3335_GPIO26_O
UART0_CTS
UART0_TX_AG3335
UART0_RX_AG3335
UART0_RTS 6
AG3335_GPIO26_O 6,91
UART0_CTS 6
UART0_TX_AG3335 4,6
UART0_RX_AG3335 4,6
J3101

蹈 笷 蒿 M T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 穝 λ カ 蒼 厩 隍 跋 縲 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title	USB			
Size D	Document Number AG3335M EVK V2			Rev 0.5
Date:	Monday, March 23, 2020	Sheet	12	of 14

Power LED Indicator


VBAT
1
J9001
2
2
R9007
R / 300 / ohm / 0402
A1
D9007
Red LED
C
BL-HGE36D-AV-TRB(CL190YG)
R9001
R / 300 / ohm / 0402
D9001
BL-HGE36D-AV-TRB(CL190YG)
R9002
R / 300 / ohm / 0402
D9002
BL-HGE36D-AV-TRB(CL190YG)
R9003
R / 300 / ohm / 0402
D9003
BL-HGE36D-AV-TRB(CL190YG)
R9004
R / 300 / ohm / 0402
D9004
BL-HGE36D-AV-TRB(CL190YG)
R9005
R / 300 / ohm / 0402
D9005
BL-HGE36D-AV-TRB(CL190YG)
R9006
R / 300 / ohm / 0402
D9006
VCCK
SSM3K36MFV
Q9001
VIO18
SSM3K36MFV
Q9002
VRF
SSM3K36MFV
Q9003
VSRAM
SSM3K36MFV
Q9004
VRTC
SSM3K36MFV
Q9005
AVDD18_TCXO
SSM3K36MFV
Q9006

Need to Check:

1. Backup_VBAT connection

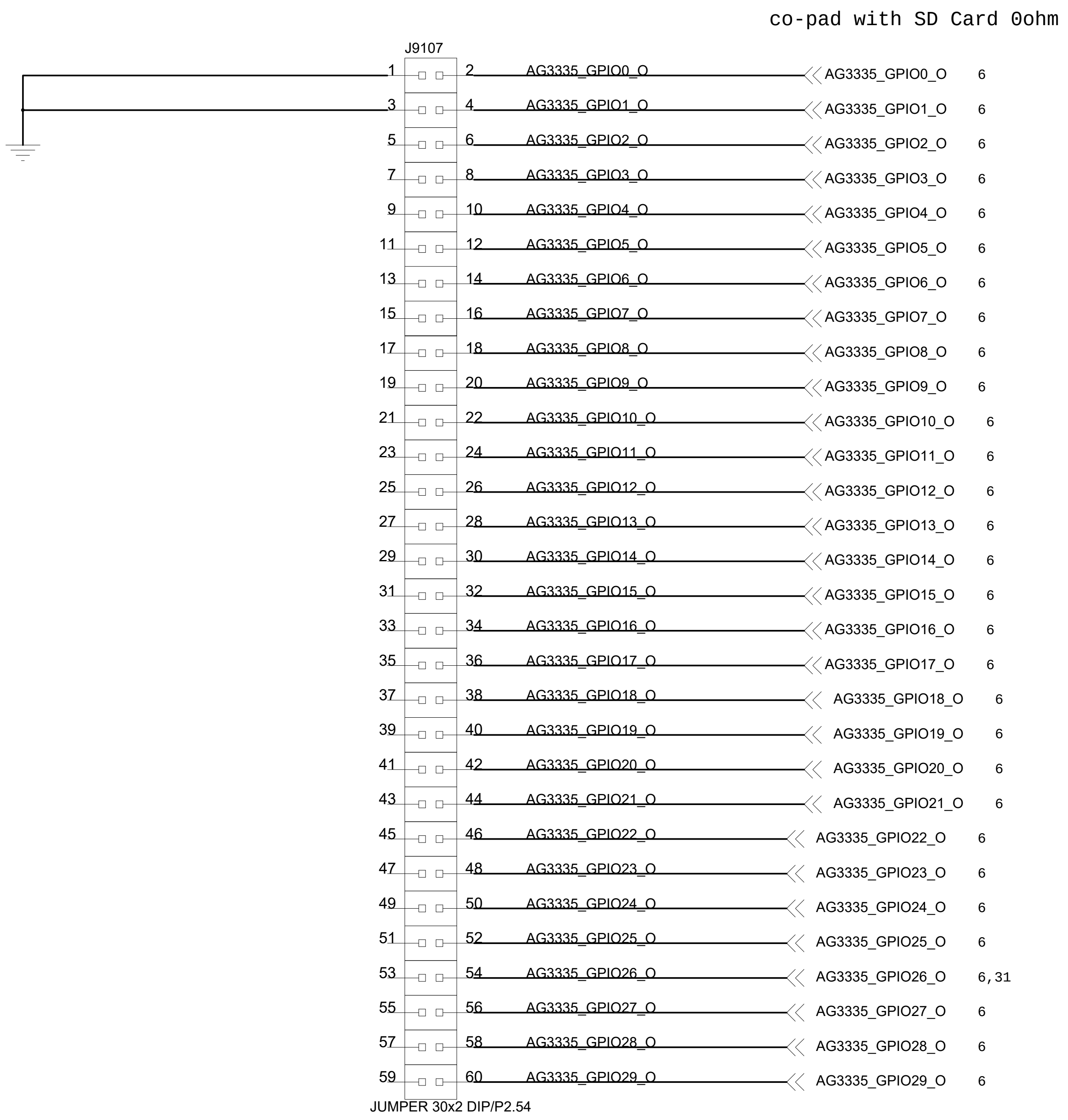
2. Low Vth NMOS Survey

踏 節 蒿 M T 綜 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 棲 入 力 荟 厩 隍 跋 縲 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title Indicator

Size D
Document Number AG3335M EVK V2
Rev 0.5

Date: Monday, March 23, 2020
Sheet 13 of 14


co-pad with SD Card 0ohm

J9107
1 2 AG3335_GPIO0_O AG3335_GPIO0_O 6
3 4 AG3335_GPIO1_O AG3335_GPIO1_O 6
5 6 AG3335_GPIO2_O AG3335_GPIO2_O 6
7 8 AG3335_GPIO3_O AG3335_GPIO3_O 6
9 10 AG3335_GPIO4_O AG3335_GPIO4_O 6
11 12 AG3335_GPIO5_O AG3335_GPIO5_O 6
13 14 AG3335_GPIO6_O AG3335_GPIO6_O 6
15 16 AG3335_GPIO7_O AG3335_GPIO7_O 6
17 18 AG3335_GPIO8_O AG3335_GPIO8_O 6
19 20 AG3335_GPIO9_O AG3335_GPIO9_O 6
21 22 AG3335_GPIO10_O AG3335_GPIO10_O 6
23 24 AG3335_GPIO11_O AG3335_GPIO11_O 6
25 26 AG3335_GPIO12_O AG3335_GPIO12_O 6
27 28 AG3335_GPIO13_O AG3335_GPIO13_O 6
29 30 AG3335_GPIO14_O AG3335_GPIO14_O 6
31 32 AG3335_GPIO15_O AG3335_GPIO15_O 6
33 34 AG3335_GPIO16_O AG3335_GPIO16_O 6
35 36 AG3335_GPIO17_O AG3335_GPIO17_O 6
37 38 AG3335_GPIO18_O AG3335_GPIO18_O 6
39 40 AG3335_GPIO19_O AG3335_GPIO19_O 6
41 42 AG3335_GPIO20_O AG3335_GPIO20_O 6
43 44 AG3335_GPIO21_O AG3335_GPIO21_O 6
45 46 AG3335_GPIO22_O AG3335_GPIO22_O 6
47 48 AG3335_GPIO23_O AG3335_GPIO23_O 6
49 50 AG3335_GPIO24_O AG3335_GPIO24_O 6
51 52 AG3335_GPIO25_O AG3335_GPIO25_O 6
53 54 AG3335_GPIO26_O AG3335_GPIO26_O 6,31
55 56 AG3335_GPIO27_O AG3335_GPIO27_O 6
57 58 AG3335_GPIO28_O AG3335_GPIO28_O 6
59 60 AG3335_GPIO29_O AG3335_GPIO29_O 6
JUMPER 30x2 DIP/P2.54

蹈 節 蒿 M T 综 そ
5F., No.6-5, Dusing Rd., Hsinchu Science Park, Hsinchu 300, Taiwan
300 穚 λ カ 斋 厫 隉 跋 縲 ︽ 隔 6-5 腹 5 加
Tel: +886-3-612-8800
Fax: +886-3-611-8833
www.airoha.com

Title GPIO

Size D
Document Number AG3335M EVK V2
Rev 0.5

Date: Monday, March 23, 2020
Sheet 14 of 14